

SILENT WAKE

Multi-Domain Naval Wargame Turn-by-Turn

Hellenic Naval academy

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Turn	Title	Facilitator	Injects	Expected and alternative play	Comment
0	Pre-sail briefing		OPS: - Mission: Independent transit to patrol sector Alpha-3. - NLT 112000B arrival. - Transit time approx. 6 hrs at planned speed. - Regional posture: Heightened readiness. - Diplomatic channels active. - Intelligence: Possible hostile naval presence in theatre. - Potential use of visual and electronic deception tactics. - No declared exclusion zones in transit corridor. - ROE: Weapons Tight. - Engagement authorized only in case of confirmed hostile intent or hostile act. - No escort assigned. - No dedicated air support during transit. - No friendly AEW coverage in corridor. - Organic helicopter joins upon arrival in patrol sector. NAV: - Navigation systems nominal. - WECDIS updated prior to departure (latest chart package + corrections). - Update completed as per standard procedure. - Route verified against latest Notices to Mariners. - Planned transit speed: 15 kts average. - Sea state 2-3. - Visibility -12 nm. - Weather clear. - Wind N 10 kts. - No shallow water constraints. - No restricted manoeuvring zones. - No anticipated navigational hazards. EO: - All primary combat, navigation, sensor, and comms systems nominal. - Air Surveillance Radar: - Minor corrective maintenance last week. - Intermittent flickering under certain clutter conditions. - Module serviced + recalibrated. - Post-maintenance checks satisfactory. - If flicker reappears -> restart affected component. - Reboot ETBOL <1 minute. - Surface tracking radar + Fire Control tested prior departure. - No defects reported. - Communications: - SATCOM functional (512 kbps allocated). - HF functional. - UHF functional. - UHF 5 occasional noise (known component issue).		Stress 0. All systems Green. Only NAV RADAR, IBMS / INS, and COMMS are ON.

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			- Replacement module pending delivery by organic helicopter. - Repair ETBOL -1 hour once part onboard. • No other open technical defects. - CyOp: • Cyber threat: Elevated. • Adversary assessed capable of: • Disruption - Data manipulation - Temporary system degradation • No specific threat indicators for this unit. • MTC experienced contained cyber incident last week. - Administrative networks affected. - Operational systems reported unaffected. - Investigation ongoing. • Shipboard networks stable. • No abnormal traffic. • No authentication anomalies. • No active alerts. • Continuous monitoring active.		

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1	In transit	You are underway. Transit phase has begun	OPS: - Transit proceeding as planned. - Estimated time to patrol sector unchanged. - No updated intelligence from Maritime Operations Center. - No declared maritime safety broadcasts affecting corridor. - EMCON: Navigation radar only NAV - Course and speed steady at 15 kts. - No deviation from planned route. - Commercial traffic density within expected levels. - ETA to patrol area: 5hours - Visibility -12 nm EO: - Radar performance nominal. - Confidential – Naval Group 6 - Silent Wake Scenario Pack - Surface and air picture stable. - No sensor anomalies observed. - Communications functioning within expected parameters. CyOp - Network traffic baseline unchanged. - No alerts. - Routine monitoring only. AAWO - Air picture normal - No contacts of concern PWO - Surface picture normal No contacts of concern	SWB01 (Continue Mission) SWB03 (Internal Sync)	
2	Subtle Anomalies		NAV: - Minor speed discrepancy between GPS SOG and STW. - One AIS contact bearing 035, range 18 nm with no corresponding radar echo. PWO: - No ESM emissions from bearing 035	SWB01 (Continue Mission Tempo) SWB07 (Start Investigation – Radar/AIS)	SOG: Speed Over Ground STW: Speed Through Water

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3	Faulty component - Increased traffic		EO: - Possible faulty component identified. Resolution: Restart (5' ETBOL). NAV: - Increase in AIS contacts within sector. - Commercial traffic density higher than expected baseline. - Majority of contacts remain beyond visual range.	SWB05 (Increase Lookouts) SWB15 (Repair Component) Alt: Contact MTC	
4	AIS spoofing possible	Component is repaired	NAV: - New AIS contact bearing 041 at 11nm. - No ship visible at this position. EO: - Component is back online.	SWB09 (AIS tracks re-evaluation) SWB01 (Continue Mission Tempo) Alt: SWB07 (Start Investigation)	First confirmed deceptive AIS track identified. Needs to be investigated. Trust in AIS picture begins to degrade. Investigation could be done to AIS
5	Cyber Alert & Comms Degradation		CyOp: - Anomaly detected in IBMS. EO: - SATCOM experiencing reduced bandwidth and intermittent connectivity. - Observed throughput below allocated 512 kbps baseline. - Confidential - Naval Group 10Silent Wake Scenario Pack - HF performance degraded. Signal clarity is inconsistent. - Communication with MTC is not possible with UHF due to distance.	SWB07 (Start Investigation - IBMS) SWB07 (Start Investigation - COMMS)	This is the first explicit cyber signal. Still ambiguous in scope. No confirmed impact yet. Communications degradation may be: - Environmental - Geographical - Network saturation - Malicious - Coincidental
6	Initial Cyber Analysis & Hostile ship spotting		CyOp: - Analysis indicates anomalous data flow originating from WECDIS towards the INS. - INS: Inertial Navigation System (responsible to feed all systems with course and speed considering GPS and inertia sensors) - Effects are not yet fully determined. NAV: - New AIS contact bearing 052, range 12 nm. - Visual silhouette indicates warship profile. PWD: - ESM detects non-commercial radar emissions at bearing 052. - Emission profile not consistent with merchant vessels.	* SWB20 (General Quarters) SWB08 (Start In-depth Investigation - IBMS)	First credible hostile identification. AIS deception is now operationally meaningful. Cyber impact chain identified but not yet fully understood. *All system are now ON (General quarters)

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7	AAW radar malfunction		AAWO: - Air surveillance radar online - Multiple tracks are flickering like last time.	SWB01 (Continue Mission Tempo) SWB14 (Reboot Affected System)	If they do not reboot now a hint will be given to the EO at next turn Reboot will trigger the silent privilege escalation in AAW radar component. No visible immediate consequence.
8	Air threat warning		AAWO: - Report via MTC (C2 system): Hostile aircrafts approaching area (ETA: 15'). EO: - Reboot could fix the AAW issue (1' ETBOL) (only if not done already)	SWB21 (Prepare Target Engagement) SWBXX (Reduce stress)	AAW radar is not operational yet. Rebooting...
9	Hostile Salvo		NAV: 3 incoming missiles visually detected. EO: - AAW radar is back online. PWD: - Target is out of gun range.	SWB29 (Decoy Deployment) *SWB27 (Fire Missile Counterattack)	WECDIS attack is not yet resolved, and that is what drives the missile counter-attack to fail Out of gun range to force them to use missiles.
10	First Blood	Defensive engagement completed. 1 missile successfully intercepted by CIWS. The other 2 were detonated in close proximity. Shockwave and fragmentation effects were observed.	EO: - FWD CIWS: non-operational (ETBOL: 2 hours) - Missiles launcher: non-operational (ETBOL: 5 hours) 1 HF antenna was disabled (unable to repair) PWD: - Target now in gun range. NAV: - Missiles (counterattack) missed the target. - No evidence of hostile manoeuvre to evade.	SWB18 (Damage Control Focus - CIWS) SWB04 (Increase Speed) SWB28 (Engage with Gun) Or SW26 (Shift to Gun Focus)	If damage control on missiles, they will never be ready for the game.

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11	Cyber Analysis & Surface Tracking Malfunction		CyOp: - Analysis indicates a cyber-attack vector resulting on feeding all system w - Option 1: Disconnect WECDIS and restart INS (ETBOL: 1') - WECDIS not available. - Option 2: Restore WECDIS and restart INS (ETBOL 30') - WECDIS available PWD: - Target within the effective gun range - Surface tracking radar is unable to achieve a stable lock. - Manual targeting reduces effective engagement envelope (reduced gun effective range)	SWB10 (Isolate System) SWB23 (Close Range to Compensate)	Isolation removes corrupted speed feed. Surface tracking instability persists due to earlier radar escalation (if reboot Closing range compensates for reduced firing geometry. Full restore (30') operationally unacceptable under air threat timeline. EO/NAV should note: Disconnecting WECDIS removes digital chart references on bridge. Manual navigation procedures can compensate for. It is not critical
12	New Cyber Alert		CyOp: - Malicious activity confirmed originating from AAW radar processing module toward - Can affect tracking capability - Originated by the "faulty" AAW radar component AAWO: - Report via MTC (C2 system): Hostile aircrafts approaching area (ETA: 10'). PWD: - Target is still out of effective gun range with manual tracking. 10' to be in range EO: - INS functioning correctly. - Speed inputs verified across systems.	SWB07 (Start Investigation – AAW radar) SWBXX – Reduce Stress	We need to emphasize the need for recovering the surface tracking. Target is too far and can hit again at any time.

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13	Hard Choices		CyOp: - AAW originated attack resolution options: - Option 1: Restore AAW radar from backup (ETBOL: 30'). Full recovery by replacing the component from storage and flashing clean firmware - Option 2: Isolate AAW radar: ETBOL < 1'. AAW radar unavailable while isolated. Surface tracking stability restored immediately. AAWO: - Report via MTC (C2 system): Hostile aircrafts approaching area (ETA: 8').	SWB10 (Isolate System (Radar)) SWB28 (Engage Target with Guns)	Isolation immediately restores surface tracking capability. Surface threat must be neutralized before incoming air threat. Air threat resolution handled after.
14	Ship neutralized - AAW threat focus		NAV: - Gun engagement achieved effective impact on hostile vessel. - Target exhibiting loss of manoeuvrability. PWD: - No further ESM emissions detected from hostile unit. - Surface threat assessed as neutralized.	SWB16 (Reconnect AAW) SWBXX (Reduce stress)	Surface threat eliminated. Air threat unresolved. AAW radar reconnection restores critical air picture.
15	Happy Ending	Hostile aircrafts were intercepted by friendly aircrafts			